

Recap:

$$y_i = \underline{\beta_0 + \beta_1 x_i} + \varepsilon_i \quad i=1, \dots, n$$

If I assume the classical SLR assumptions,

$$y_i \sim N(\underline{\beta_0 + \beta_1 x_i}, \sigma^2)$$

$$\hat{\beta}_0 \sim N(\underline{\beta_0}, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{SSX}))$$

$$\hat{\beta}_1 \sim N(\beta_1, \sigma^2/SSX)$$

Fitted values:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

What's the dist of $\hat{y}_i$?

$$\hat{y}_i \sim N(\underbrace{\beta_0 + \beta_1 x_i}_{E(\hat{y}_i)}, \underbrace{\sigma^2(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX})}_{Var(\hat{y}_i)})$$

special case
when $x_i = 0$

$$\begin{aligned} E(\hat{y}_i) &= E(\hat{\beta}_0 + \hat{\beta}_1 x_i) = E(\hat{\beta}_0) + E(\hat{\beta}_1) x_i \\ &\quad \downarrow \\ &= \beta_0 + \beta_1 x_i \end{aligned}$$

$$= Var(\hat{\beta}_0) + Var(\hat{\beta}_1 x_i) + 2Cov(\hat{\beta}_0, \hat{\beta}_1 x_i)$$

$$\text{Var}(\hat{y}_i) = \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_i)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 x_i)$$

$$= \text{Var}\left(\frac{1}{n} \sum_j y_j + \hat{\beta}_1 (x_i - \bar{x})\right)$$

$$k_j = \frac{x_j - \bar{x}}{SSX}$$

$$= \text{Var}\left(\frac{1}{n} \sum_j y_j + (x_i - \bar{x}) \sum_{j=1}^n k_j y_j\right)$$

$$= \text{Var}\left(\sum_{j=1}^n \underbrace{\left(\frac{1}{n} + (x_i - \bar{x}) k_j\right)}_{= a_j} y_j\right)$$

$$= \text{Var}\left(\sum_{j=1}^n a_j y_j\right) \quad \rightarrow \sum_{j=1}^n \text{Var}(a_j y_j)$$

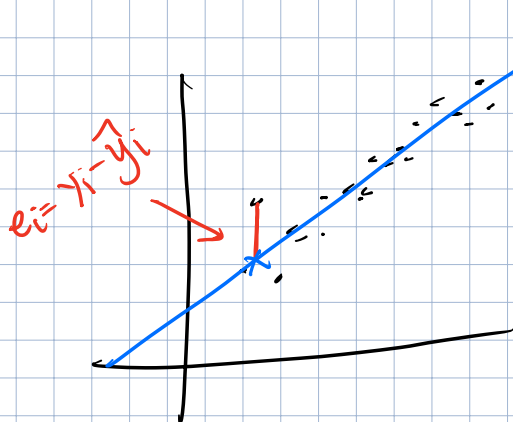
$$= \sum_{j=1}^n a_j^2 \text{Var}(y_j) = \sigma^2 \sum_{j=1}^n a_j^2$$

$$= \sigma^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{SSX} \right)$$

Residuals:

$$e_i = y_i - \hat{y}_i$$

how far off was I
for obs i



Difference b/w ϵ_i & e_i

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

$$\epsilon_i = y_i - \beta_0 - \beta_1 x_i$$

but

← never know what this is.
(not calculatable)
unknown

$$e_i = y_i - \hat{y}_i \leftarrow \text{this is calculatable}$$

$$\epsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

What is the dist of e_i ?

$$e_i \sim N\left(\boxed{0}, \boxed{\text{Hw}}\right)$$

$$\sigma^2 \left(1 - \frac{1}{n} - \frac{(x_i - \bar{x})^2}{SSX}\right)$$

(similar argument for $\text{var}(\hat{y}_i)$)

Look ahead:

$$\sum_{i=1}^n e_i = 0$$

$$e_1 + e_2 + \dots + e_n = 0$$

$$\Rightarrow e_1 = -e_2 - e_3 - \dots - e_n$$

Properties of $\hat{y}_i$ & e_i :

$$\underline{\underline{\sum_{i=1}^n e_i \sim N(0, n\sigma^2)}}$$

① $\sum_{i=1}^n e_i = 0$ (why?)

recall $\frac{\partial Q}{\partial \beta_0} \stackrel{!}{=} 0$

$$\frac{\partial}{\partial \beta_0} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0$$

So when evaluated at $(\hat{\beta}_0, \hat{\beta}_1)$ I see:

$$-2 \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\sum_{i=1}^n (y_i - \hat{y}_i) = \sum_{i=1}^n e_i = 0$$

② $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$

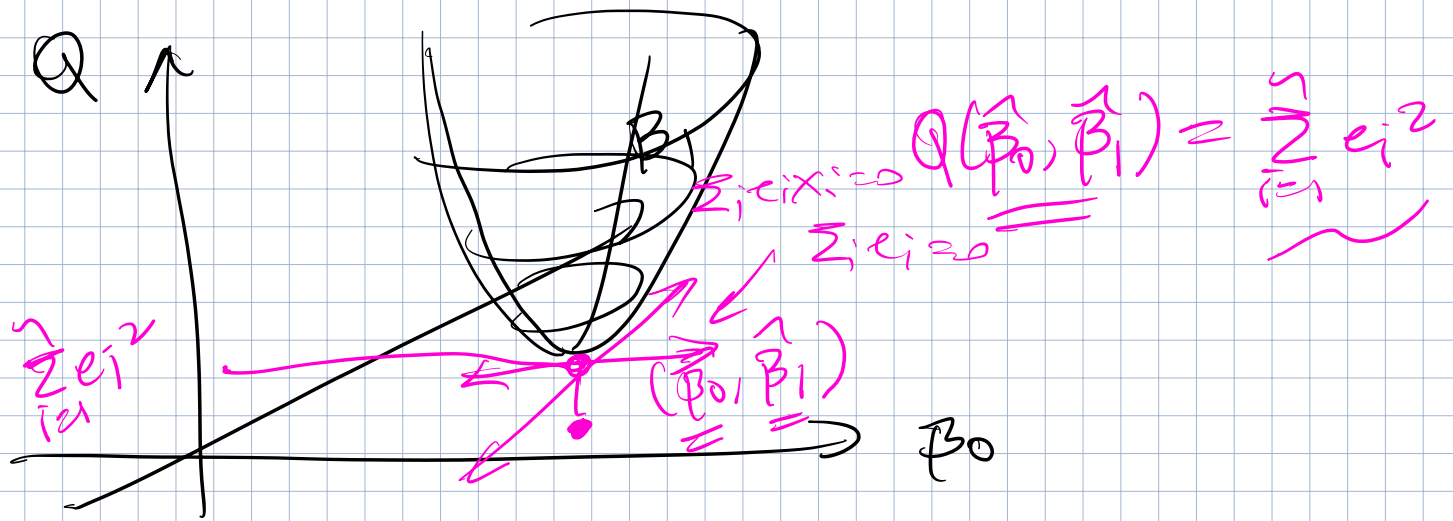
$$e_i = y_i - \hat{y}_i$$

$$\hat{y}_i = y_i - e_i$$

$$\sum_{i=1}^n \hat{y}_i = \sum_{i=1}^n y_i - \sum_{i=1}^n e_i$$

③ $\sum_{i=1}^n e_i^2$ is the global minimum of $Q(\beta_0, \beta_1)$

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$



(4) $\sum_{i=1}^n e_i x_i = 0$

consequence of $\frac{\partial Q}{\partial \beta_1} \stackrel{!}{=} 0$ at $(\hat{\beta}_0, \hat{\beta}_1)$

(5) $\sum_{i=1}^n e_i \hat{y}_i = 0$

follows by combining (1) & (4)

$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, $\varepsilon_i \stackrel{\text{ind}}{\sim} N(0, \sigma^2)$

unknown

let's say I want an unbiased estimator of σ^2 .

i.e. I'm looking for $\hat{\sigma}^2$ with $E(\hat{\sigma}^2) = \sigma^2$,

Remark

unknown

dist is exactly known

fixed

$$\frac{\sum_{i=1}^n e_i^2}{\sigma^2}$$

$\sim$

$$\chi^2_{n-p}$$

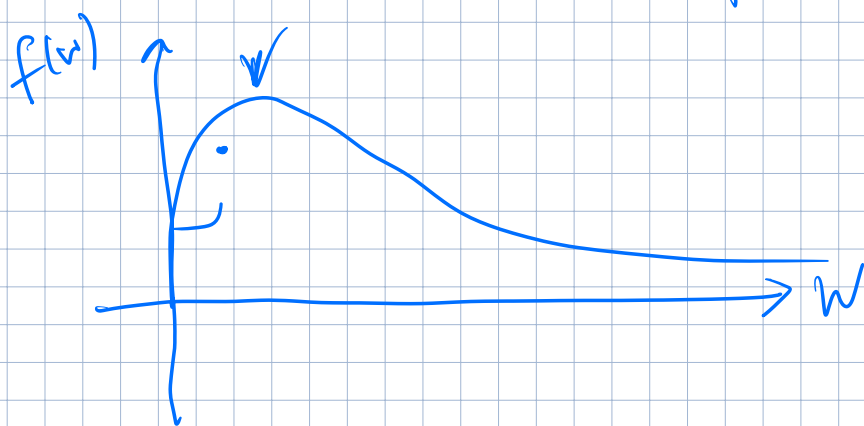
where

$p = \#$ of betas in my model

$$SLR \Leftrightarrow p=2$$

Aside:

What is a χ^2_{df} dist?



"chi-squared"

How to define/construct a χ^2_{df} RV?

11

If I had n standard normal RVs which are 11:

$\updownarrow$

"independent"

$$z_i \stackrel{iid}{\sim} N(0, 1)$$

$$V = \sum_{i=1}^n z_i^2 \sim \chi^2_n$$

"V is a chi-squared RV w/ n degrees of freedom."

Properties:

$$E(V) = n \quad \& \quad \text{Var}(V) = 2n.$$

$$\text{So } V = \frac{\sum_{i=1}^n e_i^2}{\sigma^2} \sim \chi^2_{\underline{n-2}}$$

$$E(V) = n-2$$

Goal:

$$E(\hat{\sigma}^2) = \underline{\sigma^2}$$

$$\Leftrightarrow E\left(\frac{\sum e_i^2}{\sigma^2}\right) = n-2$$

$$\Leftrightarrow \frac{1}{\sigma^2} E(\sum e_i^2) = n-2$$

$$\Leftrightarrow \frac{1}{n-2} E(\sum e_i^2) = \sigma^2$$

$$\Leftrightarrow E\left(\underbrace{\frac{\sum e_i^2}{n-2}}\right) = \sigma^2$$

call this $\hat{\sigma}^2 = \frac{\sum e_i^2}{n-2} \therefore$ it's an unbiased est of σ^2 .

$\uparrow$
MSE = mean squared error

Hypothesis Testing & Confidence Intervals for SLR

Suppose I wanted to test some claim about the true value of the slope.

We know:

$$\hat{\beta}_1 \sim N(\beta_1, \sigma^2/SSX)$$

⌈ Hypothesis testing says

$$H_0: \beta_1 = \beta_1^* \leftarrow \text{some constant}$$

$$H_1: \beta_1 \neq \beta_1^*$$

Let's try to make a statistic for this test:

$$Z = \frac{\hat{\beta}_1 - \beta_1^*}{SE(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1^*}{\sqrt{\sigma^2/SSX}}$$

$$\boxed{SE(\hat{\theta}) = \sqrt{Var(\hat{\theta})}}$$

$$= \frac{\hat{\beta}_1 - \beta_1^*}{\sigma/\sqrt{SSX}} \sim \underline{\underline{N(0, 1)}}$$

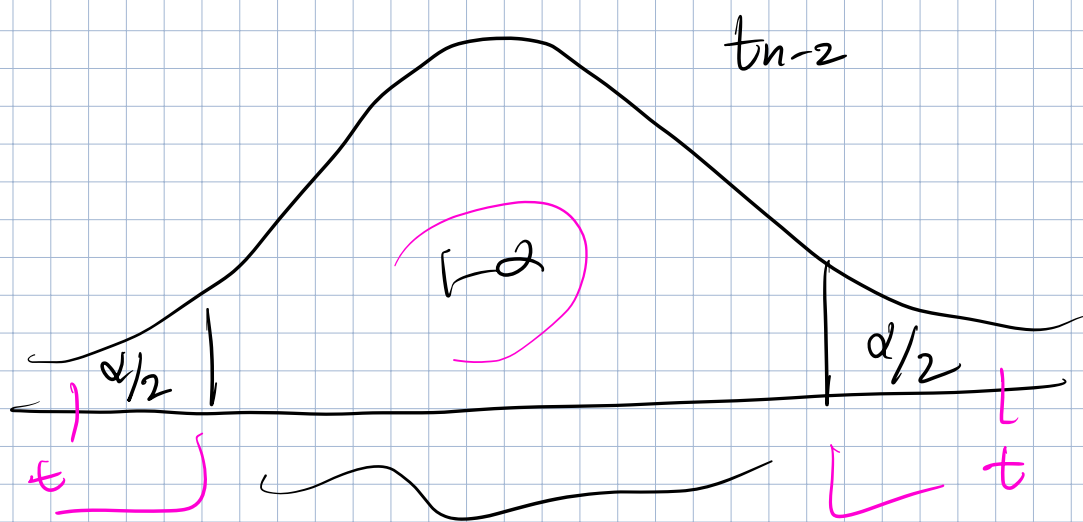
We don't know σ^2 ... if we guess it's $\hat{\sigma}^2$, we get a t-stat:

$$t = \frac{\hat{\beta}_1 - \beta_1^*}{\widehat{SE}(\hat{\beta}_1)} = \frac{\hat{\beta}_1 - \beta_1^*}{\hat{\sigma} / \sqrt{SSX}} \sim t_{n-2}$$

↑ statistic

Decision Rule:

If $|t| > t_{1-\alpha/2, n-2}^* \Rightarrow \text{reject } H_0$



unusual under null hyp.

If we find that $|t| > t_{1-\alpha/2, n-2}^*$ then we reject H_0 & have evidence that the true

slope is not actually β_1^*

When $\beta_1^* = 0$,

if we reject H_0 , we have evidence that

the slope is not 0 & so

"X is a significant predictor of Y"

$$H_0: \beta_1^* = 0 \Leftrightarrow \boxed{y_i = \beta_0 + \varepsilon_i} \quad \leftarrow X \text{ is not predictor of } Y$$
$$H_1: \beta_1^* \neq 0 \Leftrightarrow y_i = \beta_0 + \beta_1^* x_i + \varepsilon_i$$

//

Intercept:

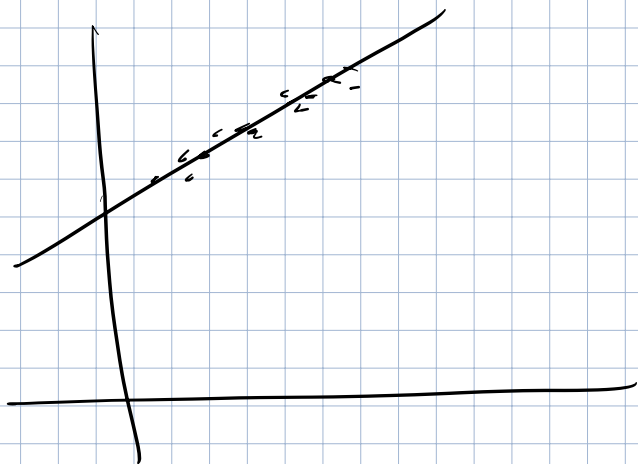
$$H_0: \beta_0 = \beta_0^*$$

$$H_1: \beta_0 \neq \beta_0^*$$

$$t = \frac{\hat{\beta}_0 - \beta_0^*}{\widehat{SE}(\hat{\beta}_0)} = \frac{\hat{\beta}_0 - \beta_0^*}{\hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{SSX}}}$$

Decision Rule:

If $|t| > t_{1-\frac{\alpha}{2}, n-2}^* \Rightarrow \text{reject } H_0$



Confidence Intervals for β_0 & β_1 :

For the slope a $100(1-\alpha)\%$ CI takes the form:

$$\hat{\beta}_1 \pm t_{1-\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_1)$$

$$\hat{\beta}_1 \pm t_{1-\alpha/2, n-2}^* \left(\hat{\sigma} / \sqrt{SSX} \right)$$

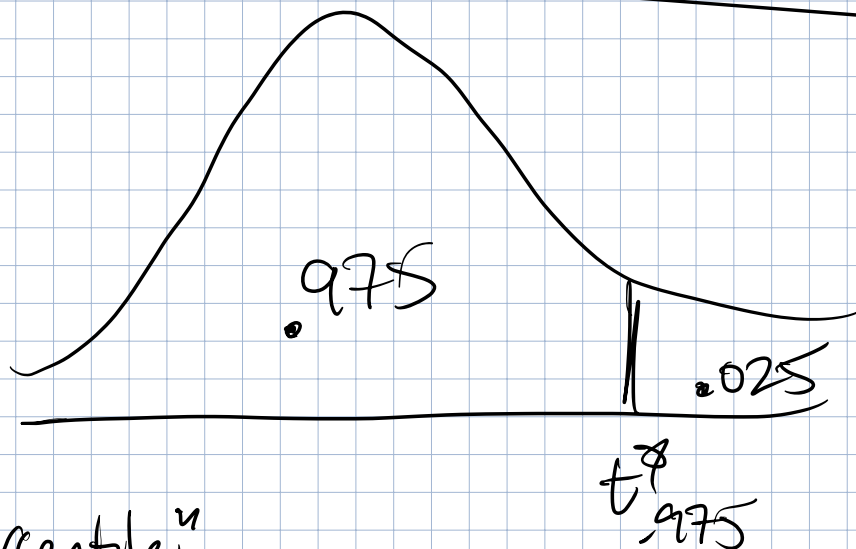
Similarly for the intercept, a $100(1-\alpha)\%$ CI is:

$$\hat{\beta}_0 \pm t_{1-\alpha/2, n-2}^* \hat{SE}(\hat{\beta}_0)$$

$$\hat{\beta}_0 \pm t_{1-\alpha/2, n-2}^* \hat{\sigma} \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{SSX}}$$

Notation

$$t_{.975}^*$$



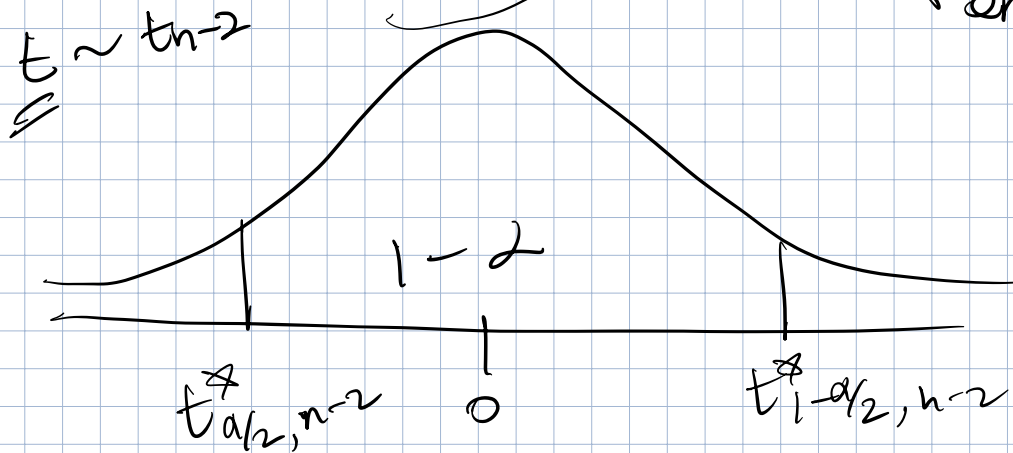
"quantile" / "percentile"

Pivot:

$$t = \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_e(\hat{\beta}_1)} \sim t_{n-2}$$

unknown parameter

entirely known dist



We know:

$$P(t_{\alpha/2, n-2} \leq \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\sqrt{SSX}} \leq t_{1-\alpha/2, n-2}) = 1-\alpha$$

$\therefore$ Solve for β_1 ← unknown parameter

$$P\left(\hat{\beta}_1 + t_{\alpha/2, n-2} \frac{\hat{\sigma}}{\sqrt{SSX}} \leq \beta_1 \leq \hat{\beta}_1 + t_{1-\alpha/2, n-2} \frac{\hat{\sigma}}{\sqrt{SSX}}\right) = 1-\alpha$$

I know by symmetry $t_{\alpha/2, n-2}^* = -t_{1-\alpha/2, n-2}^*$

$$[LB, UB]: \hat{\beta}_1 \pm t_{1-\alpha/2, n-2}^* \frac{\hat{\sigma}}{\sqrt{SSX}}$$